

DATA MINING - EC 9560

INDIVIDUAL PROJECT

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Data Pre-Processing Report

1. Project and Dataset

- Project Heading: Laptop Price Prediction using Hardware Specifications
- Dataset Used: Laptop Price Dataset (laptop_price.csv) — 1,303 laptops, 13 raw columns, no missing values
- This report covers Lab 3: transforming the raw dataset (as characterised in Lab 2) into a clean, fully numeric, model-ready form.

2. Numeric Type Conversion

Two columns were stored as text despite representing quantities and required direct suffix stripping before they could be used numerically:

- Ram: suffix "GB" removed and cast to integer → Ram_GB
- Weight: suffix "kg" removed and cast to float → Weight_kg

Both conversions were lossless — every one of the 1,303 values matched the expected "<number>GB" / "<number>kg" pattern, so no rows were dropped or coerced to null.

3. Decomposing Compound Text Columns

Memory, Cpu, Gpu, and ScreenResolution each packed multiple pieces of information into a single string, as flagged in Lab 2's closing observations. Each was parsed into separate, purpose-built features.

3.1 Memory → storage-type capacities

The Memory column held up to two storage devices per laptop joined with "+" (e.g. "256GB SSD + 1TB HDD"), with capacities in either GB or TB. Each entry was split on "+", the size and unit extracted with a regular expression (TB converted ×1000), and accumulated into four new columns by type, plus a total:

New Column	Meaning	Laptops with value > 0
SSD_GB	Solid-state storage capacity	843
HDD_GB	Hard-disk storage capacity	576
Flash_Storage_GB	Flash storage capacity	75
Hybrid_GB	Hybrid drive capacity	12
Total_Storage_GB	Sum of all storage on the device	1,303

3.2 Cpu → brand, series, and clock speed

Three features were extracted: Cpu_brand (the first token — Intel / AMD / Samsung), Cpu_type (grouping Intel chips into Core i7 / i5 / i3 / Other Intel, plus AMD Processor and Other), and Cpu_speed_GHz (the trailing "<number>GHz" pulled out with a regular expression). Distribution of Cpu_type: Intel Core i7 (527), Intel Core i5 (423), Other Intel Processor (154), Intel Core i3 (136), AMD Processor (62), Other (1).

3.3 Gpu → brand

Gpu_brand was taken as the first word of the Gpu string: Intel (722), Nvidia (400), AMD (180), ARM (1).

3.4 ScreenResolution → resolution, density, and display-type flags

The WxH pixel dimensions were extracted into Screen_Width and Screen_Height. From these, PPI (pixel density) was derived as $\sqrt{(\text{Width}^2 + \text{Height}^2)} \div \text{Inches}$ — a stronger, size-normalised measure of display quality than resolution or screen size alone. Two binary flags, Touchscreen and IPS_Panel, were set from substring checks on the original string.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Company	Type/Name	Inches	Price_euro	Item_GB	Weight_kg	SSD_GB	HDD_GB	Flash_Storage	Hybrid_GB	GE_Total	Stor_Cpu	Item_Gpu	Cpu_type	Cpu_speed	Item_Screen	W_Screen	H_PPI	Touchscreen	IPS_Panel	OpSys	Price_log	
1	Apple	Ultrabook	13.3	1339.05	0	1.37	128	0	0	128	Intel	Intel Core	2.3	Intel	2560	1600	226.99	0	1	macOS	7.20994	
2	Apple	Ultrabook	13.3	896.94	0	1.34	0	0	512	0	128	Intel	Intel Core	1.8	Intel	1440	900	127.68	0	0	macOS	6.902329
3	HP	Notebook	15.6	375	0	1.86	256	0	0	256	Intel	Intel Core	2.3	Intel	1920	1080	141.21	0	0	No OS	6.306389	
4	Apple	Ultrabook	13.4	2337.45	0	1.83	512	0	0	512	Intel	Intel Core	2.7	AMD	2880	1800	320.53	0	1	macOS	7.809099	
5	Apple	Ultrabook	13.3	1000.6	0	1.37	256	0	0	256	Intel	Intel Core	3.1	Intel	2560	1600	226.99	0	1	macOS	7.809099	
6	Acer	Notebook	15.6	400	0	2.1	0	0	300	0	300	AMD	AMD Proci	3	AMD	1366	768	100.45	0	0	Windows	5.903661
7	Apple	Ultrabook	13.4	2136.97	0	2.04	0	0	256	0	256	Intel	Intel Core	2.2	Intel	2880	1800	320.53	0	1	macOS	7.809099
8	Apple	Ultrabook	13.5	1156.7	0	1.34	0	0	256	0	256	Intel	Intel Core	1.8	Intel	1440	900	127.68	0	0	macOS	7.309817
9	Acer	Ultrabook	14	1495	0	1.3	512	0	0	512	Intel	Intel Core	1.8	Intel	1920	1080	157.35	0	1	Windows	7.21995	
10	Acer	Ultrabook	14	770	0	1.6	256	0	0	256	Intel	Intel Core	1.4	Intel	1920	1080	157.35	0	1	Windows	6.647686	
11	HP	Notebook	15.6	303.9	0	1.96	0	0	300	0	300	Intel	Intel Core	2.3	Intel	1366	768	100.45	0	0	No OS	5.978623
12	HP	Notebook	15.6	344.90	0	1.96	0	0	300	0	300	Intel	Intel Core	2	Intel	1320	1080	141.21	0	0	No OS	5.94641
13	Apple	Ultrabook	13.4	2439.97	0	1.83	256	0	0	0	256	Intel	Intel Core	2.8	AMD	2880	1800	320.53	0	1	macOS	7.809099
14	Apple	Notebook	13.8	408.3	0	2.2	256	0	0	0	256	Intel	Intel Core	2	AMD	1320	1080	141.21	0	0	Windows	6.214490
15	Apple	Ultrabook	13.2	1263.4	0	0.92	256	0	0	0	256	Intel	Other Inte	1.2	Intel	2304	1440	226.42	0	1	macOS	7.341582
16	Apple	Ultrabook	13.3	2516.50	0	1.37	256	0	0	0	256	Intel	Intel Core	2.3	Intel	2560	1600	226.99	0	1	macOS	7.39827
17	Apple	Notebook	15.6	345	0	2.2	256	0	0	0	256	Intel	Intel Core	2.7	AMD	1830	1080	141.21	0	0	Windows	6.614736
18	Apple	Ultrabook	13.4	2850	0	1.83	512	0	0	0	512	Intel	Intel Core	3.0	AMD	2880	1800	320.53	0	1	macOS	7.959277
19	Lenovo	Notebook	15.6	400	0	2.2	0	0	3000	0	1000	Intel	Intel Core	2.4	Intel	1920	1080	141.21	0	0	No OS	6.214490
20	Dell	Ultrabook	13.3	879	0	1.22	128	0	0	128	Intel	Intel Core	1.6	Intel	1920	1080	163.83	1	1	Windows	6.807593	
21	Acer	Notebook	15.6	191.9	2	0.98	0	0	32	0	32	Intel	Other Inte	1.44	Intel	1366	768	135.89	0	0	Windows	5.262172
22	Lenovo	Serving	15.6	899	0	2.5	128	0	0	1120	Intel	Intel Core	2.5	Intel	1920	1080	141.21	0	1	Windows	6.907750	
23	HP	Notebook	15.6	350	0	1.89	0	0	300	0	300	AMD	AMD Proci	1.5	AMD	1366	768	100.45	0	0	No OS	5.906608
24	Dell	2 in 1 Com	13.3	819	0	1.42	256	0	0	0	256	Intel	Intel Core	1.6	Intel	1920	1080	163.83	1	0	Windows	6.709098
25	HP	Ultrabook	15.6	430	0	1.91	256	0	0	0	256	Intel	Intel Core	1.8	Intel	1920	1080	141.21	0	0	Windows	6.803097
26	Dell	Notebook	15.6	418.54	0	2.3	0	0	3000	0	1000	Intel	Intel Core	2	Intel	1366	768	100.45	0	0	Windows	6.030977
27	Apple	Ultrabook	13.5	1009	0	1.35	0	0	128	0	128	Intel	Intel Core	1.6	Intel	1440	900	127.68	0	0	macOS	7.309099
28	Dell	Notebook	15.6	800	0	2.2	256	0	0	0	256	Intel	Intel Core	1.6	AMD	1320	1080	141.21	0	0	Windows	6.809661
29	Dell	Ultrabook	15.6	1296	0	1.88	512	0	0	0	512	Intel	Intel Core	1.8	Intel	1920	1080	141.21	0	0	Windows	7.16839
30	HP	Notebook	17.3	896	0	2.0	0	0	3000	0	1000	Intel	Intel Core	1.6	Intel	1920	1080	127.34	0	0	Windows	6.709096
31	Chival	Notebook	15.6	344.90	0	1.88	0	0	64	0	64	Intel	Other Inte	1.44	Intel	1920	1080	141.21	0	0	Windows	5.905291
32	Acer	Notebook	14	290	2	1.83	0	0	32	0	32	AMD	AMD Proci	1.5	AMD	1366	768	111.94	0	0	Windows	5.268217

Figure 1: laptop_price_engineered.csv

4. Categorical Grouping

OpSys was consolidated by merging "Mac OS X" into "macOS" (matching Lab 2's combined Apple-OS count of 21), producing OpSys_grouped as the categorical field carried forward for encoding.

5. Retained Inches and Weight_kg

Lab 2 noted that Inches and Weight_kg are highly collinear ($r = 0.828$) and each has a weak individual correlation with price ($r = 0.068$ and $r = 0.210$ respectively). Both columns were nonetheless kept and fully processed in this stage rather than removed. Preprocessing (Lab 3) is a cleaning and feature-construction step; deciding which features to eliminate on statistical grounds — with a documented method such as a correlation threshold, Variance Inflation Factor, or Recursive Feature Elimination — is the explicit purpose of the Lab 4 feature-selection stage. Removing a feature here would pre-empt that analysis rather than support it.

6. Target Variable: Correcting Skew

Lab 2 measured Price_euros skewness at ≈ 1.52 (right-skewed). A log1p transform was applied to produce Price_log, retained alongside the original Price_euros so the model can be trained on whichever the chosen algorithm handles better.

Version	Skewness
Price_euros (raw)	1.521
Price_log = log(1 + Price_euros)	-0.172

7. Encoding Categorical Variables

Company, TypeName, OpSys_grouped, Cpu_brand, Cpu_type, and Gpu_brand were converted using One-Hot Encoding (with the first category of each dropped to avoid the dummy-variable trap), since none of these are ordinal — there is no inherent ranking between, say, Dell and Lenovo. This expanded the dataset from 22 engineered columns to 56 fully numeric columns.

8. Feature Scaling

All continuous numeric features (Inches, Ram_GB, Weight_kg, the four storage columns, Total_Storage_GB, Cpu_speed_GHz, Screen_Width, Screen_Height, PPI) were standardised with StandardScaler (zero mean, unit variance). This matters for distance- and gradient-based algorithms (e.g. KNN, SVR, regularised linear regression) that would otherwise let large-magnitude features such as Screen_Width dominate small-magnitude ones such as Weight_kg. The binary flags (Touchscreen, IPS_Panel) and one-hot columns were left unscaled, as scaling 0/1 indicators has no meaningful effect.

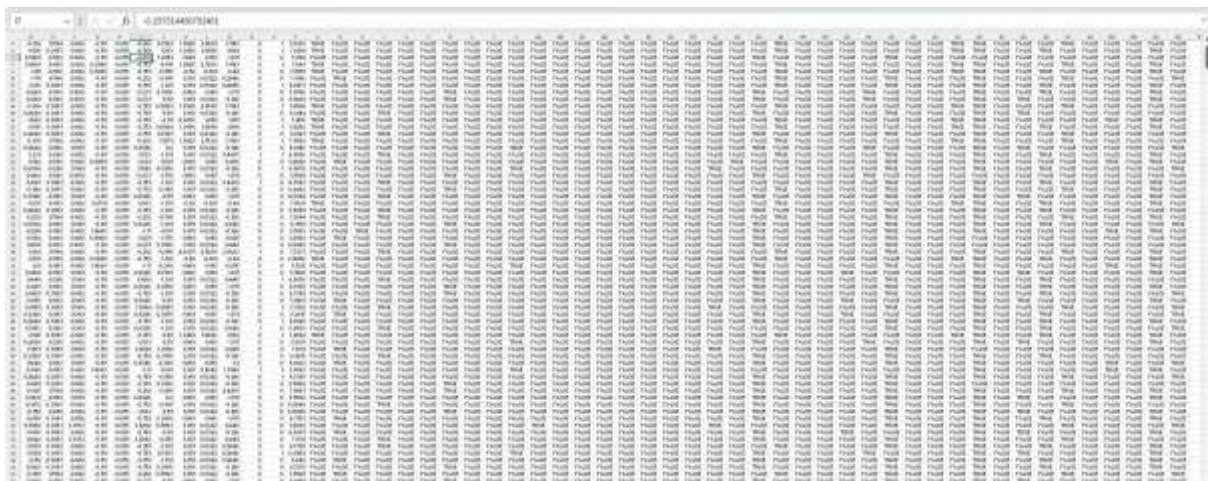


Figure 2:laptop_price_finale.csv

9. Validating the New Features

10. Final Dataset Summary

Stage	Rows	Columns	Missing values
Raw (laptop_price.csv)	1,303	13	0
Engineered (laptop_price_engineered.csv)	1,303	22	0
Model-ready, encoded + scaled (laptop_price_model_ready.csv)	1,303	56	0